

Cross-Modal Fusion for Tamil Sign Language Vowel Recognition from RGB Video and Flex Sensor Data

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Abstract

Automatic recognition of Tamil Sign Language (TSL) remains challenging because visually similar gestures may differ in subtle finger articulations, while RGB-based systems are sensitive to background clutter, occlusions, and lighting changes. We present a multimodal framework for isolated recognition of 13 one-handed TSL vowel signs by combining RGB video and flex sensor signals collected from a custom dataset of 30 signers. The study compares image-only, sensor-only, and multimodal models under a fixed signer-independent evaluation setting. We consider CNN baselines for video and sensor streams, and then introduce a transformer-based multimodal architecture with bidirectional cross-attention, modality gating, and an auxiliary alignment objective. On the test split, the sensor-based transformer achieves 80.0% accuracy, while the best multimodal configuration reaches 73.9%, clearly improving over image-only modeling. These results indicate that flex sensor signals provide the strongest discriminative information in the current setup, while multimodal fusion offers a viable framework for low-resource sign language recognition.

1. Introduction

Communication between deaf and hard-of-hearing individuals and the broader population is still limited by the low diffusion of sign language literacy in daily social, educational, and professional settings. Automatic sign language recognition (SLR) aims to reduce this gap by translating gestures into textual or spoken output.

Vision-based SLR has benefited from recent advances in deep learning, where convolutional and transformer-based models process RGB images or videos to recognize hand gestures. These approaches have shown strong performance, but they remain sensitive to illumination changes, background clutter, occlusions, and viewpoint variation. In addition, camera-based systems may raise privacy concerns in real deployment scenarios.

Wearable sensor-based systems offer a complementary perspective by directly measuring hand articulation through flex or inertial sensors. These systems are generally more robust to environmental noise and occlusion, but their adoption is limited by hardware constraints, the scarcity of annotated datasets, and the limited study of multimodal learning strategies. These issues are even more evident for Tamil Sign Language (TSL), where benchmark datasets and well-established recognition pipelines are still limited.

In earlier work, a smart glove with five flex sensors was proposed for isolated recognition of 13 Tamil vowel signs using a one-dimensional convolutional neural network trained only on sensor readings, with real-time text-to-speech output [1]. While that study showed the feasibility of sensor-based TSL vowel recognition, it did not include visual context, signer-independent evaluation, or explicit multimodal fusion.

In this paper, we study isolated Tamil Sign Language vowel recognition from synchronized RGB video and flex sensor data. We compare unimodal baselines and multimodal fusion strategies, including a late-fusion model and a transformer-based cross-attention architecture. The goal is to assess whether explicit interaction between visual and sensor representations improves robustness and cross-

signer generalization.

Contributions.

- We introduce a multimodal TSL benchmark for 13 one-handed vowel gestures, combining synchronized RGB video clips and flex sensor sequences collected from 30 signers
- We evaluate image-only, sensor-only, and multimodal architectures, including CNN baselines and a transformer-based cross-attention model with modality gating
- We provide an ablation study on multimodal alignment and show that progressive scheduling is important to avoid early over-regularization

2. Related Work

2.1. Vision-Based Sign Language Recognition

Vision-based sign language recognition has progressed substantially with the availability of large-scale corpora and deep sequence models. RWTH-PHOENIX-Weather established one of the first widely used large-vocabulary benchmarks for sign language recognition and translation [4]. Later, Camgoz et al. introduced neural sign language translation, extending the field from isolated recognition toward end-to-end sequence modeling from video [3]. Li et al. further strengthened word-level visual recognition through the large-scale WLASL dataset [6]. More recently, SignBERT showed that self-supervised hand-model-aware pretraining can improve downstream sign language recognition performance [5].

2.2. Sensor-Based Sign Language Recognition

Wearable sensing provides a complementary alternative to purely vision-based systems, especially in the presence of occlusions or uncontrolled environments. Prior work has explored glove-based and wearable sign language recognition using flex sensors and inertial signals [13]. More recently, Lu et al. combined bending-sensor data with visual skeletal information and deep temporal modeling, showing that multimodal sensor fusion can improve robustness under occlusion compared with using either modality alone [7].

2.3. Multimodal and Radar-Based Sign Language Recognition

Multimodal learning has become increasingly relevant for robust sign language recognition. In the medical communication domain, Mineo et al. introduced a multimedia/multimodal dataset for patient-doctor interaction, combining RADAR, LiDAR, RGB, and RGB-D sensing [9]. Caligiore et al. further studied multisource isolated LIS recognition on the MultiMedaLIS dataset, highlighting the utility of combining visual modalities and structured motion information [2]. Radar-based sign language recognition has also emerged as a privacy-aware alternative to camera-

based approaches. Mineo et al. proposed a radar imaging framework for LIS recognition in medical communication [10], while TRACE introduced text-aligned contrastive learning for radar-based sign language recognition [11].

2.4. Challenges and Benchmarks

Recent benchmarks have emphasized the importance of signer-independent evaluation and multimodal robustness. AUTSL introduced a large-scale Turkish sign language benchmark with explicit user-independent assessment [14]. SignEval 2025 further highlighted these challenges in both continuous and isolated settings, including a multimodal isolated sign recognition track based on MultiMedaLIS [8]. In the radar-specific setting, benchmark extensions have started to standardize evaluation for privacy-aware isolated sign language recognition [12].

3. Methodology

3.1. Dataset Creation

We collected a multimodal dataset for isolated recognition of 13 one-handed Tamil Sign Language vowel gestures. Each acquisition consists of one RGB sequence and one synchronized flex sensor sequence, and every sample is associated with a signer identifier and a gesture label. After pairing the visual and sensor streams, discarding incomplete acquisitions, and retaining only samples with reliable temporal alignment across the two modalities, the final benchmark used in the experiments contains 312 paired samples, of which 247 are used for training and 65 for testing. The split is signer-disjoint, i.e., no signer appears in both training and test sets. The test set contains five samples per class and is kept fixed throughout the study.

RGB data are acquired with a camera while participants perform the gestures under varying illumination and background conditions. In parallel, five resistive flex sensors mounted on the fingers capture bending patterns throughout the gesture. Sensor readings are acquired through an Arduino Mega2560 at approximately 20 Hz using a 50 ms sampling interval, converted through the onboard 10-bit ADC, and stored as CSV files. Each sensor sequence contains five channels and is standardized to a temporal length of 160 steps.

Video capture and sensor logging are started simultaneously through a shared acquisition script. After recording, both modalities are trimmed to the gesture interval and, when needed, padded or truncated to 160 temporal steps. This fixed-length representation makes the data compatible with both CNN-based and transformer-based pipelines.

During acquisition, participants are instructed to perform each gesture with one hand in a natural position while minimizing unrelated motion. The resulting dataset captures variability in hand size, articulation speed, and gesture ex-

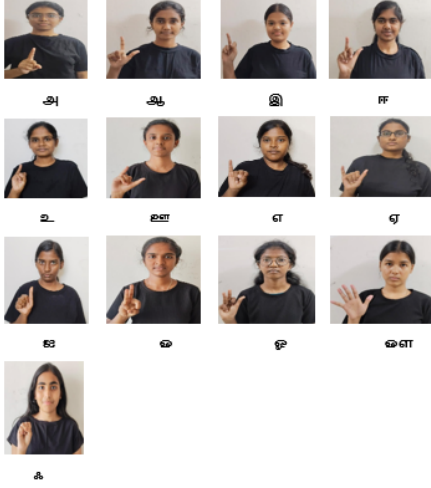


Figure 1. Examples of the 13 Tamil vowel signs across different signers

ecution while preserving the temporal correspondence required for multimodal learning.

We formulate sign recognition as a 13-class classification problem. For the CNN-based visual model, a sample is an RGB clip

$$X_I \in \mathbb{R}^{3 \times T \times H \times W},$$

while the sensor stream is represented as

$$X_S \in \mathbb{R}^{5 \times T},$$

where $T = 160$. The target label is

$$y \in \{1, \dots, 13\}.$$

3.2. Image-Based Baseline (3D-CNN)

The image-only baseline uses a 3D convolutional neural network to encode RGB clips. Each input consists of 160 RGB frames resized to 112×112 . The network contains three convolutional blocks with 32, 64, and 128 output channels, respectively. Each block includes a 3D convolution, batch normalization, ReLU activation, and pooling. An adaptive average pooling layer is then applied to obtain a compact visual representation

$$f_I \in \mathbb{R}^{128}.$$

This feature vector is passed to a linear classifier that predicts one of the 13 gesture classes.

3.3. Sensor-Based Baseline (1D-CNN)

The sensor-only baseline processes flex sensor signals with a 1D-CNN. Raw sensor readings are preprocessed through

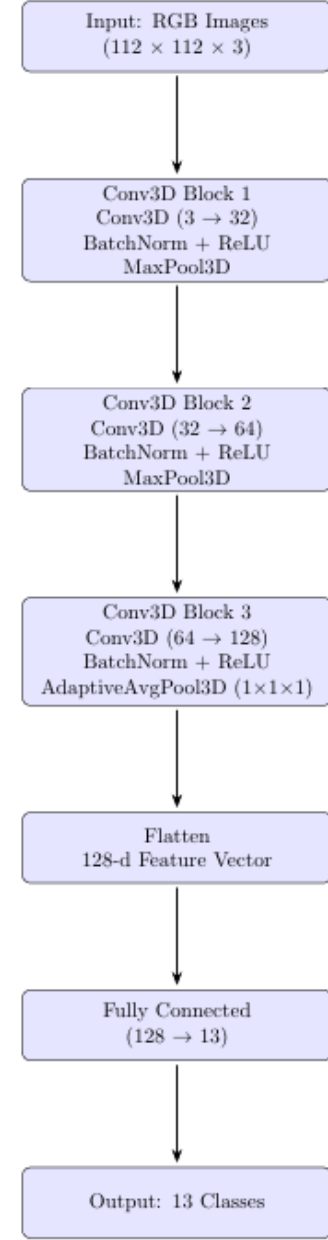


Figure 2. Image-based baseline based on a 3D-CNN

moving-average smoothing and Min-Max normalization. The resulting input is represented as

$$X_S \in \mathbb{R}^{5 \times 160}.$$

The network consists of two temporal convolutional blocks that map the input channels from 5 to 32 and then from 32 to 64. Each block includes a 1D convolution with kernel size 3 and padding, followed by batch normalization and ReLU activation. Adaptive average pooling produces a

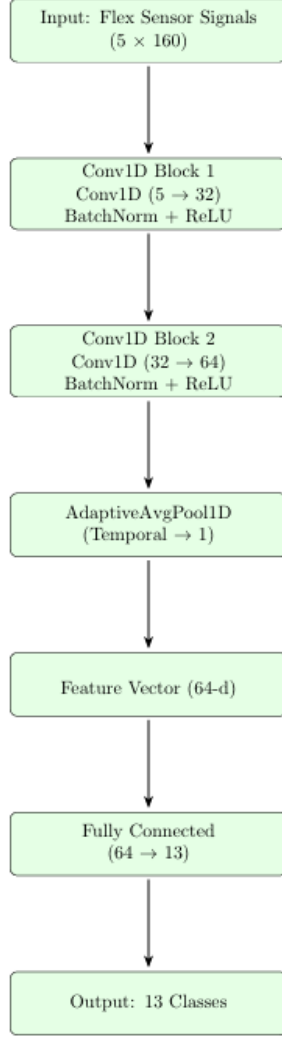


Figure 3. Sensor-based baseline based on a 1D-CNN

compact sensor representation

$$f_S \in \mathbb{R}^{64},$$

which is fed to a fully connected classifier for 13-way prediction.

3.4. Late-Fusion Baseline

As a first multimodal baseline, we adopt late fusion. The visual feature vector $f_I \in \mathbb{R}^{128}$ and the sensor feature vector $f_S \in \mathbb{R}^{64}$ are concatenated to obtain a joint representation

$$f = [f_I; f_S] \in \mathbb{R}^{192}.$$

The fused vector is processed by two fully connected layers. The first maps 192 features to 128 dimensions and is followed by ReLU and dropout with rate 0.4. The final classification layer outputs scores for the 13 gesture classes.

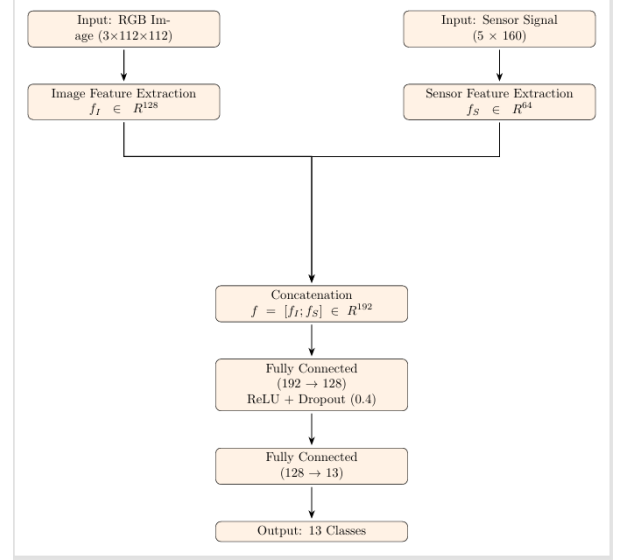


Figure 4. Late-fusion baseline combining visual and sensor features

This design is straightforward and computationally efficient, but it does not model explicit interactions between the two modalities before the final classifier.

3.5. Transformer-Based Multimodal Model

To model cross-modal dependencies more explicitly, we design a transformer-based multimodal architecture. The visual and sensor streams are encoded separately and fused through bidirectional cross-attention.

For the visual stream, we use a pretrained ViT-Small/16-224 backbone. Since this branch operates on image tokens rather than spatiotemporal volumes, it receives the central RGB frame of each sequence, resized to 224×224 and normalized with ImageNet statistics. The image is partitioned into 16×16 patches, producing 196 patch tokens. A learnable CLS token and positional embeddings are added before the transformer encoder. To reduce overfitting on a compact dataset, the first six transformer blocks are frozen and the remaining layers are fine-tuned. The resulting visual representation is projected to a shared embedding space with dimension $d_{\text{model}} = 256$.

The sensor stream is modeled as a multichannel sequence

$$X_S \in \mathbb{R}^{5 \times 160}.$$

A 1D convolution with kernel size 8 and stride 4 is used to generate temporal tokens:

$$N_p = \left\lfloor \frac{160 - 8}{4} \right\rfloor + 1 = 39.$$

Each token is projected to the same 256-dimensional space. A learnable CLS token and sinusoidal positional encoding

are added, followed by dropout. The token sequence is then processed by a lightweight transformer encoder.

The two token sequences are fused through bidirectional cross-attention. In one direction, image tokens attend to sensor tokens; in the other, sensor tokens attend to image tokens. This allows the model to align visual appearance with finger-bending dynamics. The attention operator is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

Residual connections, layer normalization, and feed-forward blocks are used after each attention module.

To reduce the impact of noisy or weak modalities, we introduce a lightweight gating mechanism after cross-attention. Let z_I and z_S denote the modality-level CLS embeddings. We compute

$$[\alpha_I, \alpha_S] = \text{softmax}(\text{MLP}([z_I; z_S])),$$

and build the final multimodal descriptor as

$$z_f = [\alpha_I z_I; \alpha_S z_S].$$

The fused feature is passed to a classification head composed of a fully connected layer, GELU, dropout, and a final linear layer for 13-class prediction.

To encourage consistency across modalities, we also use an auxiliary alignment loss between the visual and sensor CLS tokens:

$$L_{\text{align}} = 1 - \frac{\langle z_I, z_S \rangle}{\|z_I\| \|z_S\|}.$$

3.6. Training Strategy

The baseline CNN models are trained with the Adam optimizer and cross-entropy loss for multi-class classification. We use a batch size of 32 and train for 50 epochs. A step-wise learning rate scheduler reduces the learning rate by a factor of 0.5 every 5 epochs.

The transformer-based models are optimized with AdamW, learning rate 1×10^{-4} , weight decay 1×10^{-4} , and cosine annealing scheduling. The main objective is cross-entropy loss. For multimodal fusion, we also evaluate an auxiliary alignment term:

$$L = L_{CE} + \lambda L_{\text{align}}. \quad (1)$$

We consider four fusion configurations: no alignment, fixed alignment weight, reduced alignment weight, and reduced alignment weight with progressive scheduling. In the scheduled setting, the alignment term is introduced after a short warm-up and increased gradually, which prevents the alignment objective from dominating the early optimization phase.

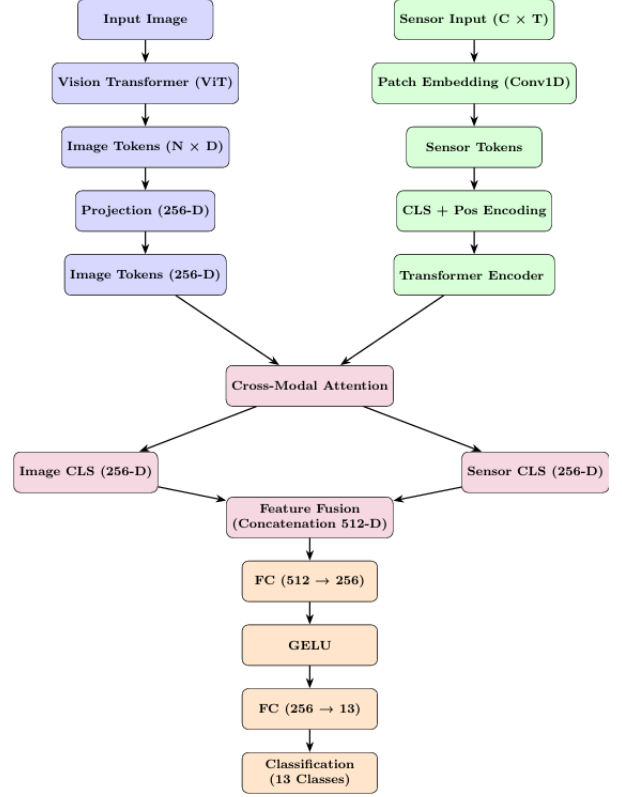


Figure 5. Transformer-based multimodal architecture with bidirectional cross-attention and modality gating

4. Experimental Methodology and Evaluation

4.1. Protocol and Metrics

All experiments are conducted on the same fixed train/test split composed of 247 training samples and 65 test samples. Performance is reported using classification accuracy, macro-averaged precision, macro-averaged recall, and macro-averaged F1-score. In addition to aggregate metrics, we examine confusion patterns and summarize class-wise and signer-wise behavior in the discussion.

For the CNN pipeline, the visual branch receives clips of 160 frames resized to 112×112 , while the sensor branch receives five normalized channels padded or truncated to 160 temporal steps. For the transformer pipeline, the image branch receives one representative frame per sequence resized to 224×224 , and the sensor branch uses the same normalized 5-channel input with temporal length 160. All models are trained for 50 epochs with batch size 32.

4.2. Main Results

Table 1 compares the CNN baselines and the transformer-based models on the fixed test split. Three main trends emerge. First, sensor signals are substantially more informative than the visual branch in the current setup. Second,

Table 1. Comparison of unimodal and multimodal models on the fixed test split. All metrics are reported as percentages (%).

Model	Accuracy	Precision	Recall	F1-score
3D-CNN (RGB only)	33.85	35.14	33.85	33.18
1D-CNN (Sensor only)	72.31	73.46	72.31	72.49
Late Fusion CNN	69.23	70.15	69.23	68.89
Image Transformer	7.69	9.29	7.69	7.67
Sensor Transformer	80.00	81.47	80.00	80.22
Fusion Transformer	73.85	74.21	73.85	73.65

Table 2. Ablation study on alignment for transformer fusion. All metrics are reported as percentages (%).

Configuration	Accuracy	F1-score
No alignment	73.85	73.65
Fixed alignment weight	72.31	72.11
Reduced alignment ($\lambda = 0.01$)	72.31	71.88
Reduced alignment + warm-up/ramp	73.85	73.65

multimodal fusion consistently improves over image-only modeling. Third, the best overall performance is obtained by the sensor-only transformer, while the fusion transformer remains the strongest multimodal configuration.

The 3D-CNN substantially outperforms the image-based transformer in the current setup. This result should be interpreted in light of the different input protocols: the CNN processes the full temporal clip, whereas the visual transformer relies on a single representative frame. Even under this caveat, the large gap between the image-only and sensor-only transformer suggests that finger-bending trajectories carry a substantial portion of the discriminative information for these isolated TSL vowels.

At the multimodal level, both late fusion and transformer fusion markedly improve over the image-based transformer, and the fusion transformer also surpasses the late-fusion CNN baseline. This indicates that explicit cross-modal interaction and gated fusion provide a more effective integration strategy than simple feature concatenation. At the same time, the fusion transformer does not exceed the sensor-only transformer, suggesting that the visual branch, in its current form, does not yet contribute enough complementary information to improve over the strongest sensor-driven model.

4.3. Ablation on Alignment

We next analyze the effect of the auxiliary alignment objective on the transformer-based fusion model. Table 2 reports the main ablation results.

The ablation shows that the alignment objective must be

introduced carefully. When applied from the beginning with a fixed or unscheduled weight, it slightly degrades classification performance, suggesting that the model becomes over-constrained during the early stages of optimization. By contrast, warm-up and ramp scheduling stabilize training and recover the same performance as the no-alignment setting. In the current configuration, alignment does not provide an absolute gain, but progressive scheduling prevents the degradation observed with fixed alignment and makes the auxiliary constraint compatible with the main classification objective.

5. Discussion

The main finding of this study is the dominance of the sensor signal. In the current setup, the sensor-only transformer is the strongest model by a clear margin, reaching 80.00% accuracy and 80.22% macro F1-score. This result suggests that flex sensor trajectories encode the most reliable class-discriminative information for isolated TSL vowel gestures.

The second finding is that multimodal fusion is useful but not yet optimal. Both multimodal pipelines improve substantially over purely image-based modeling, which confirms that combining appearance information and articulation dynamics is a sound direction. The transformer-based fusion model is also stronger than the late-fusion CNN baseline, supporting the benefit of explicit cross-modal interaction. However, fusion still does not surpass the sensor-only transformer. This indicates that the visual branch, as currently configured, does not yet provide sufficiently strong complementary cues to improve over the most informative sensor-driven representation.

The third finding concerns the alignment objective. The underlying rationale is appropriate, since image and sensor embeddings should occupy compatible regions of a shared feature space. However, enforcing this constraint too early interferes with the classification objective. Progressive scheduling addresses this issue and yields more stable optimization behavior, but without producing a net improvement over the no-alignment configuration.

The class-wise and signer-wise analysis, not shown for brevity, is consistent with these trends. The most challeng-

ing cases correspond to gestures with similar dynamic patterns and limited visual separability. The sensor branch remains the most stable across signers, while the fusion model shows moderate sensitivity to signer variability, largely inherited from the visual stream. Overall, these observations support a clear conclusion: the proposed multimodal framework is coherent and promising, but the sensor modality currently drives performance.

6. Conclusion

We presented a multimodal framework for isolated Tamil Sign Language vowel recognition from RGB video and flex sensor data. The study compares CNN baselines and transformer-based models on a fixed benchmark covering 13 gesture classes. The experiments show that flex sensor signals are the strongest source of discriminative information in the current setting, while multimodal fusion remains competitive and clearly improves over image-only modeling. The ablation study further shows that multimodal alignment should be introduced progressively, since unscheduled alignment can degrade classification performance. These findings position the proposed framework as a solid basis for future work on stronger visual encoders, more effective multimodal integration, and larger signer-diverse TSL benchmarks.

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